Homothety and centres of similitude

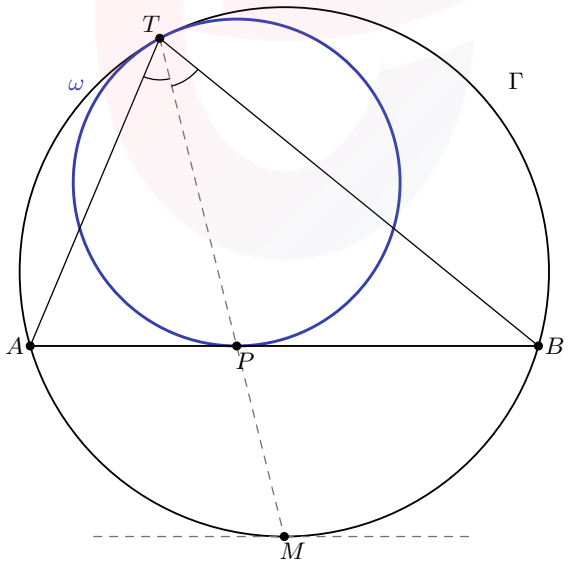
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L4-X6 Level 4 Olympiad · Geometry · Beyond the textbook

Homothety as a proof tool, covering centres of similitude, touching circles, Monge's theorem, the incircle and excircle lemma and the nine-point circle.
By the end you can find the homothety in a figure with touching circles, parallel lines or circles in an angle; use centres of similitude and Monge's theorem to prove collinearity and concurrency; use the incircle and excircle lemma and the nine-point circle as homothetic images.
Prerequisites: Olympiad angle chasing (L4-GE1), Lengths and ratios (L4-GE2). **Contests:** BMO1, MOG, AIME, USAJMO. **Time:** about 120 minutes.

1 Lesson

Circle ω lies inside circle Γ (centre O) and touches Γ at T. A chord AB of Γ touches ω at P. Prove that TP bisects $\angle ATB$. Enlarge ω from T into Γ. The scale factor exceeds 1, so the tangent AB goes to a parallel tangent of Γ farther from T, and P goes to its contact point M. So T, P, M are collinear, and $OM \perp AB$ makes M the midpoint of arc AB not containing T.



This is GCSE enlargement, called *homothety* in olympiads. Textbooks redo each such fact with fresh similar triangles, but one map does the work of all of them. The three-circle theorem is Monge's, and Euler found the line OGH in 1765.

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1.1 Homothety

Theorem 1.
The homothety with centre S and ratio $k \neq 0$ sends X to X' with $\overrightarrow{SX'} = k\,\overrightarrow{SX}$.
(1) $\overrightarrow{X'Y'} = k\,\overrightarrow{XY}$, so lines go to parallel lines, lengths scale by $|k|$, angles are kept.
(2) The circle with centre P and radius r goes to the circle with centre P' and radius $|k|r$, and tangency is kept.
(3) Ratio k_1 at S_1 followed by ratio k_2 at S_2, with $k_1k_2 \neq 1$, is a homothety with ratio k_1k_2 and centre on line S_1S_2.

Proof. (1) $\overrightarrow{X'Y'} = \overrightarrow{SY'} - \overrightarrow{SX'}$, and (2) follows. (3) The composite equals $X \mapsto Z + k_1k_2(X - Z)$ for

$$Z = \frac{(1-k_2)S_2 + k_2(1-k_1)S_1}{1-k_1k_2},$$

whose weights add up to 1.

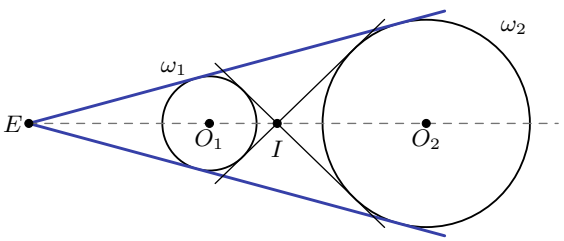
Theorem 2 (spotting a homothety).
If $\overrightarrow{A'B'} = k\,\overrightarrow{AB}$ with $k \neq 1$, a homothety with ratio k sends $A \mapsto A'$, $B \mapsto B'$, so AA' and BB' pass through its centre. Hence if triangles ABC, $A'B'C'$ have corresponding sides parallel and are not translates, AA', BB', CC' are concurrent.

Proof. Take $S = \frac{A'-kA}{1-k}$. Then $S + k(B-S) = A' + k(B-A) = B'$. For triangles, $k\,\overrightarrow{AB} + k'\,\overrightarrow{BC} \parallel \overrightarrow{AB} + \overrightarrow{BC}$ forces $k = k'$.

1.2 Two circles

Theorem 3 (centres of similitude).
Let ω_1, ω_2 have centres $O_1 \neq O_2$ and radii r_1, r_2. The homotheties taking ω_1 to ω_2 have ratios r_2/r_1 (if $r_1 \neq r_2$) and $-r_2/r_1$, and their centres E and I are the *external* and *internal centres*, divide O_1O_2 externally and internally in the ratio $r_1 : r_2$.
(1) Common external tangents pass through E, internal ones through I.
(2) If the circles touch at T, then $T = I$ for external touching and $T = E$ for internal.

Proof. The map sends $O_1 \mapsto O_2$ and scales radii by $|k|$, so $k = \pm r_2/r_1$ and $\overrightarrow{SO_2} = k\,\overrightarrow{SO_1}$ fixes S. (1) For a common tangent touching at T_1, T_2, $\overrightarrow{O_2T_2} = \pm\frac{r_2}{r_1}\overrightarrow{O_1T_1}$ (+ if the circles are on the same side), so apply Theorem 2. (2) T is on O_1O_2 with $TO_1 = r_1$, $TO_2 = r_2$.



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Theorem 4 (Monge).
For three circles with distinct centres and pairwise different radii, the three external centres are collinear, and the internal centres of two pairs are collinear with the external centre of the third.

Proof. The homothety at E_{12} then at E_{23} takes ω_1 to ω_3 with ratio $\frac{r_3}{r_1} \neq 1$. By Theorem 1(3) the centre of this map is on $E_{12}E_{23}$, and by Theorem 3 the centre is E_{13}. With I_{12}, I_{23} the ratio is $\left(-\frac{r_2}{r_1}\right)\left(-\frac{r_3}{r_2}\right) = \frac{r_3}{r_1}$ again.

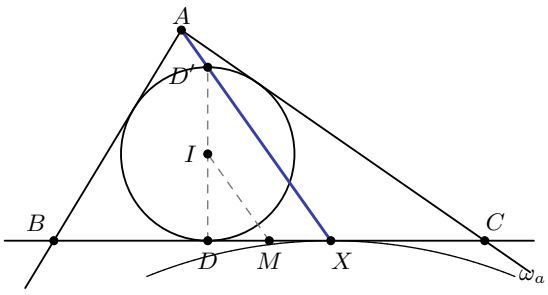
Remark (Desargues). If triangles ABC and $A'B'C'$ have $AB \parallel A'B'$, $BC \parallel B'C'$ and $CA \parallel C'A'$, a homothety or a translation takes one to the other, so AA', BB', CC' are concurrent or parallel. This is Desargues' theorem with the axis of perspective at infinity. The general theorem, two triangles perspective from a point are perspective from a line, is the projective form of the same statement, and Monge's theorem follows from Desargues in the same way.

Remark (Miquel). A spiral similarity is a homothety followed by a rotation about the same centre. The Miquel point of a complete quadrilateral is the centre of the spiral similarity taking one pair of opposite sides to the other, so the arguments of this unit carry over to L5-GE2.

1.3 The incircle and the excircle

Lemma 5.
The incircle of ABC (centre I, radius r) touches BC at D, and DD' is a diameter. The A-excircle (centre I_a, radius r_a) touches BC at X. Then
(1) A, D', X are collinear and $AD' : AX = r : r_a$,
(2) $BX = CD = s - c$, so D, X are symmetric about the midpoint M of BC, and
(3) $MI \parallel AX$.

Proof. The homothety h at A with ratio r_a/r fixes lines AB, AC, so h takes the incircle to the excircle. With n the unit normal to BC towards A, $D' = I + rn$ and $X = I_a + r_a n = h(D')$. For (2), if the excircle touches AB, AC at Z, Y, then $AY + AZ = (c + BX) + (b + CX) = 2s$, so $AZ = s$ and $BX = BZ = s - c = CD$. (3) is the midline of triangle $DD'X$.



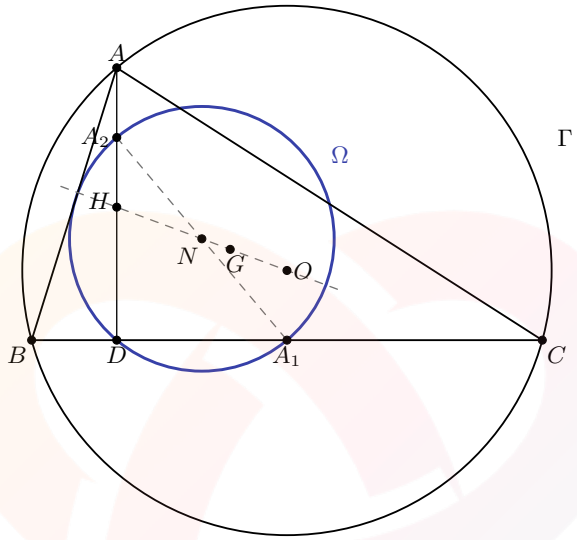
1.4 The nine-point circle

Theorem 6.
Let ABC have circumcircle Γ (centre O, radius R), centroid G, orthocentre H, and let Ω be the image of Γ under the homothety at H with ratio $\frac{1}{2}$.
(1) Ω passes through the midpoints of the sides, the feet of the altitudes and the midpoints

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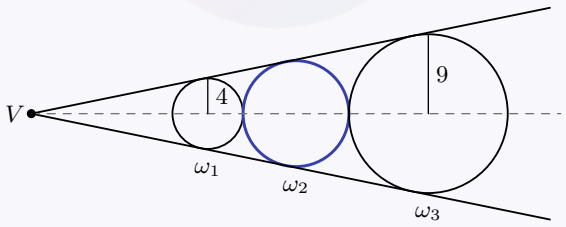
of AH, BH, CH, and its centre N is the midpoint of OH.
(2) The homothety at G with ratio $-\frac{1}{2}$ also takes Γ to Ω, sending A, B, C to the midpoints of the opposite sides and O to N.
(3) H, G, N, O are collinear (the Euler line), with $HG = 2\,GO$.

Proof. With origin O, $h = a + b + c$ satisfies $(h - a) \cdot (b - c) = |b|^2 - |c|^2 = 0$, so $h = H$, and $G = h/3$ gives (3). The map $X \mapsto \frac{X+h}{2}$ sends a to the midpoint A_2 of AH and $-a$ to the midpoint A_1 of BC, so A_1A_2 is a diameter of Ω, and $\angle A_2DA_1 = 90^\circ$ puts the foot D on Ω. The map $X \mapsto \frac{h-X}{2}$ sends $a \mapsto A_1$ and $0 \mapsto N$. ■



1.5 Worked examples

Example 1.
Circles $\omega_1, \omega_2, \omega_3$ lie inside an angle with vertex V, each touching both arms, and ω_2 touches ω_1 and ω_3, and ω_1 is nearest to V. The radii of ω_1 and ω_3 are 4 and 9. Find the radius of ω_2.



Solution.
The standard way is to take half-angle θ and radius x, the tangencies give

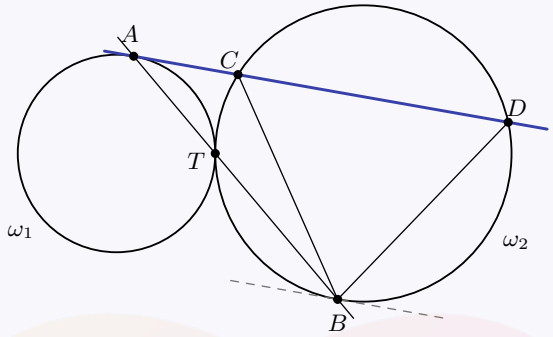
$$\frac{x-4}{\sin\theta} = x + 4, \qquad \frac{9-x}{\sin\theta} = 9 + x.$$

Dividing, $2x^2 = 72$ and $x = 6$.
With a homothety, note that V is the external centre of each pair. The homothety at V with ratio $x/4$ takes ω_1 to ω_2, hence ω_2 to the inscribed circle touching ω_2 beyond ω_2, which is ω_3. So $9 = x \cdot \frac{x}{4}$ and $x = 6$.

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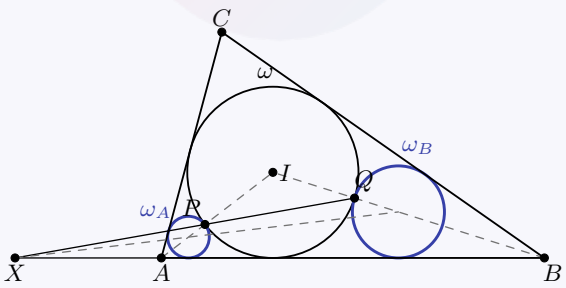
Remark. Radii of a chain of circles in an angle form a geometric progression, and the angle never enters.

Example 2.
Circles ω_1 and ω_2 touch externally at T. A line through T meets ω_1 again at A and ω_2 again at B. The tangent to ω_1 at A meets ω_2 at C and D. Prove that $BC = BD$.



Solution.
The homothety at T taking ω_1 to ω_2 sends A to the other point of ω_2 on line TA, which is B, and the tangent at A to the tangent at B. So CD is parallel to the tangent at B, hence perpendicular to O_2B, where O_2 is the centre of ω_2. Thus O_2B is the perpendicular bisector of chord CD, and $BC = BD$. ■
Remark. With touching circles, try the homothety at the contact point first.

Example 3.
Triangle ABC has incircle ω with centre I and radius r, and $CA \neq CB$. The circle ω_A inside the triangle touches AB, AC and touches ω externally, and ω_B is defined in the same way at B. Segments AI and BI meet ω at P and Q. Prove that line PQ, line AB and the line through the centres of ω_A and ω_B are concurrent.



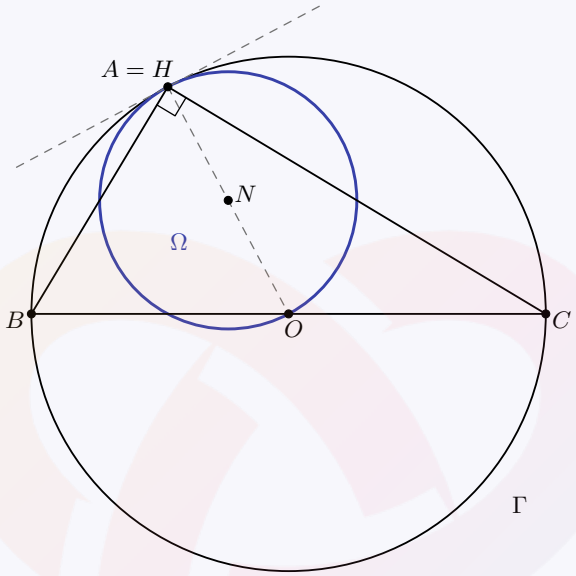
Solution.
By Theorem 3, A and P are the external and internal centres of ω_A and ω. Similarly B and Q for ω_B and ω.
As $\angle A \neq \angle B$, the radii differ, since the homothety at A taking ω to ω_A sends the far end of the diameter of ω on AI to P, so

$$\frac{\rho_A}{r} = \frac{AI - r}{AI + r} = \frac{1 - \sin\frac{A}{2}}{1 + \sin\frac{A}{2}},$$

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which decreases in A.
So the external centre X of ω_A and ω_B, which is on their line of centres, lies on AB and on PQ by Monge's theorem for ω_A, ω_B, ω. ■
Remark. Menelaus would need the radii exactly, but here we only needed them unequal.

Example 4.
Prove that the nine-point circle of a triangle touches its circumcircle if and only if the triangle has a right angle.



Solution.
If the triangle is equilateral the circles are concentric. Otherwise, if Γ and Ω touch at T, then T is a centre of similitude, H or G. As G is inside Γ, $T = H$ lies on Γ.
With origin O, $|a+b+c|^2 = R^2$ gives $a\cdot b + b\cdot c + c\cdot a = -R^2$, so $(a+b)\cdot(a+c) = 0$, that is $OC_1 \perp OB_1$ for the midpoints C_1, B_1 of AB, AC. If $C_1 = O$ or $B_1 = O$, a side is a diameter, and otherwise $AB \perp AC$.
Conversely, if $\angle A = 90^\circ$ then $H = A$ lies on Γ and on Ω, and the homothety at H sends the tangent to Γ at H to a parallel tangent to Ω through H, which is the same line. ■

Common traps.

- Circles touching externally have the contact point as internal centre, ratio negative.
- Equal radii give no external centre (a translation), so check before using Monge.
- Two pairs of parallel sides do not make triangles homothetic.
- In a proof, state centre, ratio and what goes to what, because a line may meet the image circle twice, so justify each image point.

1.6 How to spot a homothety

1. Touching circles suggest the contact point.

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2. Parallel segments, or a tangent parallel to a chord, suggest Theorem 2.
3. Circles in one angle suggest the vertex, and a diameter of the incircle suggests Lemma 5.
4. Midpoints of sides or of AH, BH, CH suggest Theorem 6.
5. Three circles suggest Monge.

2 Practice problems

The problems are roughly in order of difficulty, from ⋆⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆⋆ Triangle ABC has circumcircle Ω and incentre I. A circle ω inside Ω lies on the same side of BC as A, touches segment BC at E and touches Ω at a point $T \neq A$. Prove that the circumcircle of triangle TEI is tangent to line AI.

Problem 2 ⋆⋆⋆ Circles ω_1 and ω_2 have different radii and meet at A and B. One common external tangent touches ω_1 at P and ω_2 at Q. Line AP meets ω_2 again at Y, and line BQ meets ω_1 again at Z. Prove that line YZ passes through the point where the two common external tangents of ω_1 and ω_2 meet.

Problem 3 ⋆⋆⋆ Triangle ABC has $AB \neq AC$. By Feuerbach's theorem, which you may assume, the nine-point circle of ABC touches the incircle internally at K and touches the excircles opposite A, B, C externally at K_a, K_b, K_c. Lines KK_a and K_bK_c meet line BC at P and Q. Prove that $\angle PAQ = 90^\circ$.

Problem 4 ⋆⋆⋆⋆ Triangle ABC has $AB \neq AC$, and M is the midpoint of BC. The incircle ω touches BC, CA, AB at D, E, F, and the A-excircle touches BC, CA, AB at X, Y, Z. The tangent from M to ω other than BC touches ω at K. Prove that K lies on the circumcircles of triangles XEY and XFZ.

Problem 5 ⋆⋆⋆⋆ Triangle ABC has incircle ω, which touches BC, CA, AB at D, E, F, and I_a is the centre of the A-excircle. A circle Γ passes through B and C and touches ω at N. Prove that N lies on the circumcircles of triangles BFI_a and CEI_a.

Problem 6 ⋆⋆⋆⋆ Circles ω_1 and ω_2 touch each other externally at T, and touch a circle Γ internally at A and B. A common external tangent of ω_1 and ω_2 meets Γ at P and Q. The common tangent of ω_1 and ω_2 at T meets Γ at U and V, where V is on the same side of line PQ as T. Prove that T is the incentre of triangle PQV.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **BMO1 2012/13 Problem 2** ⋆⋆
   Look at the homothety at X taking S to T and see where the homothety sends A.

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2. **IMO 1978 Problem 4** ⋆⋆
   Compare the circle with the circumcircle via the homothety at their tangency point, and watch where the arc midpoint goes.

3. **IMO 1981 Problem 5** ⋆⋆
   Locate the centres of the three circles on the angle bisectors and compare their triangle with ABC.

4. **BMO1 2011/12 Problem 3** ⋆⋆⋆
   Use the homothety at X taking S to S1 and the one at Y taking S to S2. Where do P-related points go?

5. **AIME 2007 II Problem 15** ⋆⋆⋆
   The centres of the three equal circles and the incentre are related by a homothety, so find its ratio and then use the incircle radius.

6. **Baltic Way 2011 Problem 14 (via imogeometry.blogspot.com Baltic Way index)** ⋆⋆⋆
   Use the diameter FG of the incircle and the homothety at C (or the excircle one) that carries the incircle to a circle tangent to AB.

7. **Baltic Way 2007 Problem 15** ⋆⋆⋆
   Look at the homothety at B between the incircle and the second circle, and where D goes.

8. **Baltic Way 1992 Problem 19** ⋆⋆⋆
   Send each small circle to C by its homothety at A or B and track the tangent line t.

9. **BMO2 2015/16 Problem 1** ⋆⋆⋆
   For two touching circles and a common tangent, express the distance between the touch points in terms of the radii.

10. **AIME I 2019 Problem 15** ⋆⋆⋆⋆
    Homothety at each tangency point sends the small circle to omega, and combine with the radical axis PQ and power of P.

11. **IMO Shortlist 1999 G6** ⋆⋆⋆⋆
    Use the homothety at M taking $Omega_1$ to Omega and see what happens to the chord CD and to the line AB.

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4 Hints

1. Where does the homothety at T send E? Compare the power of that point with its distance from I.

2. Show that $\angle PBQ + \angle PAQ = 180°$. Then compare arc PZ of ω_1 with arc QY of ω_2.

3. Every point of contact is a centre of similitude. Which centre of similitude of the incircle and the A-excircle lies on BC? And of the B- and C-excircles?

4. First prove that K lies on line AX, using the circle with centre M through D. Then compute $AK \cdot AX$.

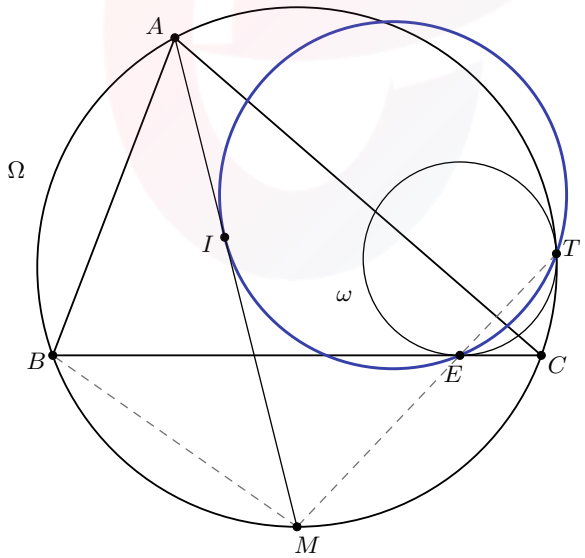
5. The homothety at N taking ω to Γ sends D to the midpoint W of an arc BC of Γ. Show that W is the midpoint of DI_a.

6. Send the contact points of PQ with ω_1 and ω_2 to Γ by the homotheties at A and B. Then find the power of the image point with respect to both small circles.

5 Solutions

Problem 1

Triangle ABC has circumcircle Ω and incentre I. A circle ω inside Ω lies on the same side of BC as A, touches segment BC at E and touches Ω at a point $T \neq A$. Prove that the circumcircle of triangle TEI is tangent to line AI.



Let M be the midpoint of the arc BC of Ω not containing A, so M lies on AI. The homothety h at T taking ω to Ω has ratio greater than 1. Line BC touches ω at E with ω on the side of A, so $h(BC)$ is the tangent to Ω parallel to BC with Ω on the side of A. This tangent touches Ω at M, so $h(E) = M$, and T, E, M are collinear with E between T and M.

Next, $\angle MBE = \angle MBC = \angle MAC = \frac{1}{2}\angle A$, and $\angle MTB = \angle MAB = \frac{1}{2}\angle A$, as T and A both lie on arc BAC, away from arc BM. So triangles MBE and MTB, which share the angle at M, are similar, and $MB^2 = ME \cdot MT$.

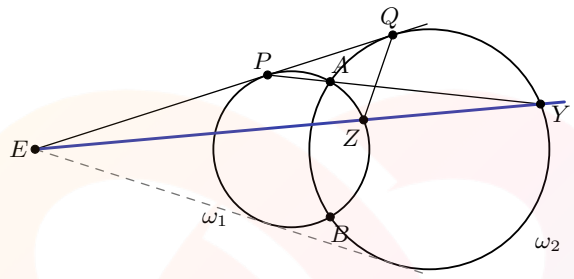
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By the incentre-excentre lemma $MI = MB$, so $MI^2 = ME \cdot MT$. Line TM meets Ω only at T and M, and $T \neq A$, so line TM is not line AM and $I \notin TM$. As M lies on line TE outside segment TE, the power of M with respect to circle TEI is $ME \cdot MT = MI^2$. So line MI meets circle TEI only at I, and touches the circle there. Line MI is line AI. ■

Remark. Only the contacts with BC and Ω were used. So for every such circle ω, the circle TEI touches AI at the same point I.

Problem 2

Circles ω_1 and ω_2 have different radii and meet at A and B. One common external tangent touches ω_1 at P and ω_2 at Q. Line AP meets ω_2 again at Y, and line BQ meets ω_1 again at Z. Prove that line YZ passes through the point where the two common external tangents of ω_1 and ω_2 meet.



Let O_1, O_2 be the centres, $r_1 \neq r_2$ the radii and E the external centre. By Theorem 3, both external tangents pass through E, and the homothety h at E with ratio $k = r_2/r_1$ takes ω_1 to ω_2 and P to Q. We use directed angles $\measuredangle$ modulo 180° as in L4-GE1.

Line PQ touches ω_1 at P and ω_2 at Q, so the tangent-chord angle gives

$$\measuredangle(BP, BA) = \measuredangle(PQ, PA), \qquad \measuredangle(BA, BQ) = \measuredangle(QA, QP).$$

Adding, $\measuredangle(BP, BQ) = \measuredangle(QA, PA) = \measuredangle(AQ, AP)$. Without directions this says $\angle PBQ + \angle PAQ = 180^\circ$.

Now Z is on line BQ and Y is on line AP. By the inscribed angle theorem, the rotation about O_1 taking P to Z has angle $2\measuredangle(BP, BQ)$ modulo 360°, and the rotation about O_2 taking Q to Y has angle $2\measuredangle(AQ, AP)$. (If $Z = B$, line BQ is the tangent at B and the same holds, and similarly if $Y = A$.) These two angles are equal. Vectors $\overrightarrow{O_1P}$ and $\overrightarrow{O_2Q}$ point the same way, as both are perpendicular to PQ and the circles lie on the same side of PQ. Hence $\overrightarrow{O_1Z}$ and $\overrightarrow{O_2Y}$ point the same way too, so $\overrightarrow{O_2Y} = k\overrightarrow{O_1Z}$, which says $h(Z) = Y$.

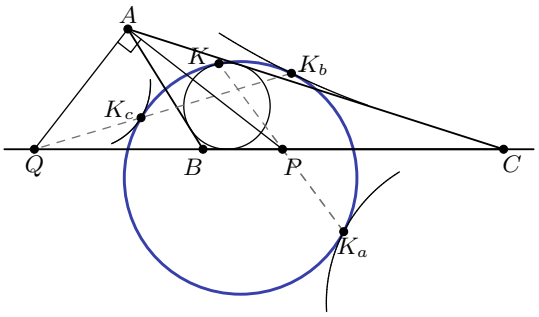
So E, Z, Y are collinear. Also $Y \neq Z$, since otherwise $h(Z) = Z$ forces $Z = E$, a point of line PQ on ω_1, so $Z = P$ and $Q = h(P) = P$, which is impossible for circles meeting at two points. ■

Remark. Showing $E \in YZ$ is the same as showing $Y = h(Z)$, and this reduces to comparing two arcs.

Problem 3

Triangle ABC has $AB \neq AC$. By Feuerbach's theorem, which you may assume, the nine-point circle of ABC touches the incircle internally at K and touches the excircles opposite A, B, C externally at K_a, K_b, K_c. Lines KK_a and K_bK_c meet line BC at P and Q. Prove that $\angle PAQ = 90^\circ$.

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Let the incircle ω and the excircles ω_a, ω_b, ω_c have centres I, I_a, I_b, I_c and radii r, r_a, r_b, r_c, and let the nine-point circle Ω have radius ρ. Let AI meet BC at L, and let the external bisector at A meet BC at L', which exists as $AB \neq AC$. We show that $P = L$ and $Q = L'$. Then $\angle PAQ = 90°$, as the two bisectors at A are perpendicular.

Centres on BC. Circles ω and ω_a touch BC from opposite sides, so BC is a common internal tangent and passes through their internal centre, which lies on line $II_a = AI$. So the homothety at L with ratio $-r_a/r$ takes ω to ω_a. Circles ω_b and ω_c both lie on the side of BC containing A, so BC is a common external tangent, and their centres lie on the external bisector at A. So their external centre is L'. This centre exists because $r_b \neq r_c$. Indeed, splitting at I_b gives $[ABC] = [ABI_b] + [CBI_b] - [ACI_b] = r_b(s-b)$, similarly $[ABC] = r_c(s-c)$, and $s - b \neq s - c$.

Contacts as centres. By Theorem 3(2), K is the external centre of ω and Ω, and K_a, K_b, K_c are internal centres. Also $\rho \neq r$, since two circles touching internally with equal radii coincide, while the midpoint M of BC is on Ω but not on the incircle (the incircle meets BC only at D, and $D \neq M$ by Lemma 5(2)).

Monge. The homothety at L with ratio $-r_a/r$ followed by the one at K_a with ratio $-\rho/r_a$ takes ω to Ω with ratio $\rho/r \neq 1$. By Theorem 1(3) the composite is a homothety with centre on line LK_a, and the only homothety with positive ratio taking ω to Ω has centre K. So L lies on line KK_a. In the same way, the homothety at K_b taking ω_b to Ω followed by the one at K_c taking Ω to ω_c has ratio $\left(-\frac{\rho}{r_b}\right)\left(-\frac{r_c}{\rho}\right) = \frac{r_c}{r_b} \neq 1$, so the external centre L' of ω_b and ω_c lies on line K_bK_c.

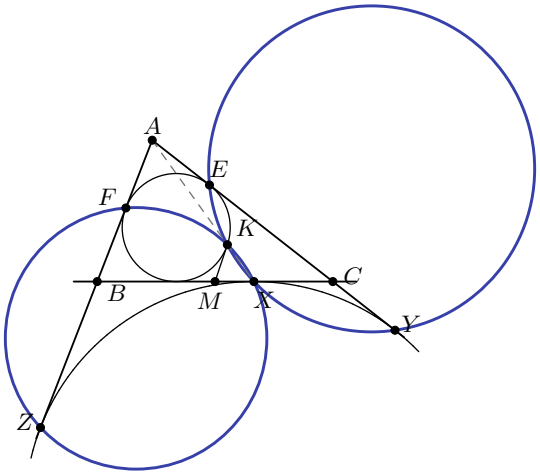
Neither line is BC. The nine-point circle meets BC only at M and at the foot H_a of the altitude from A. If K, K_a were both on BC, they would be the contact points D, X of ω, ω_a with BC. These are distinct with midpoint M, so both would equal H_a, a contradiction. Likewise ω_b and ω_c meet BC only beyond C and beyond B, away from M, so K_b, K_c cannot both lie on BC. Hence $P = L$ and $Q = L'$. ■

Remark. This is the proof of Monge's theorem. We redo the proof instead of quoting Theorem 4 because r_a may equal ρ.

Problem 4

Triangle ABC has $AB \neq AC$, and M is the midpoint of BC. The incircle ω touches BC, CA, AB at D, E, F, and the A-excircle touches BC, CA, AB at X, Y, Z. The tangent from M to ω other than BC touches ω at K. Prove that K lies on the circumcircles of triangles XEY and XFZ.

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Write a, b, c for the sides and s for the semiperimeter. Tangent lengths give $AE = AF = s - a$ and $AY = AZ = s$ (proof of Lemma 5), and Lemma 5(2) gives $MD = MX$. As $AB \neq AC$, $D \neq X$.

Step 1. Equal tangents give $MK = MD = MX$, so K lies on the circle with diameter DX and $\angle DKX = 90°$. Here $K \neq X$, since X is a point of the tangent BC other than D and so lies outside ω. Let DD' be a diameter of ω. Then $K \neq D'$, because $MD'^2 = MD^2 + DD'^2 > MK^2$, so $\angle DKD' = 90°$ as well. Thus D' and X both lie on the line through K perpendicular to KD. By Lemma 5(1), A, D', X are collinear with $AD' : AX = r : r_a$. So K is the second point where line AX meets ω, and D', K lie on ray AX.

Step 2. The power of A with respect to ω gives $AD' \cdot AK = AE^2 = (s-a)^2$. Splitting the area at I and at I_a gives $[ABC] = rs = r_a(s-a)$, so $AX = \frac{r_a}{r} AD' = \frac{s}{s-a} AD'$. Hence

$$AK \cdot AX = \frac{s}{s-a}\, AK \cdot AD' = s(s-a) = AE \cdot AY.$$

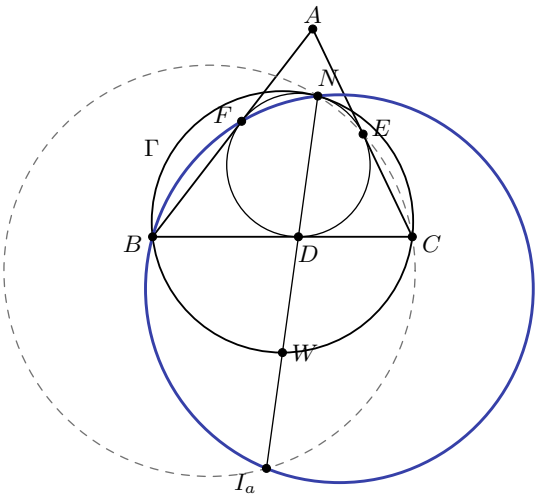
Points K, X lie on one ray from A and E, Y on another, so by the converse of power of a point, X, K, E, Y are concyclic. The same computation with $AF \cdot AZ = s(s-a)$ puts K on circle XFZ. ■

Remark. Step 1 is the hard step. The tangent from M seems to have nothing to do with A, and the diameter DD' is what links K to Lemma 5.

Problem 5

Triangle ABC has incircle ω, which touches BC, CA, AB at D, E, F, and I_a is the centre of the A-excircle. A circle Γ passes through B and C and touches ω at N. Prove that N lies on the circumcircles of triangles BFI_a and CEI_a.

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Set-up. Point D is inside segment BC, a chord of Γ, so D is inside Γ. If ω and Γ touched externally, no point of ω would be inside Γ, and Γ cannot lie inside ω since B is outside ω. So ω lies inside Γ, $N \neq D$, and the homothety h at N taking ω to Γ has ratio greater than 1.

Step 1. Line BC touches ω at D with ω on the side of A, so $h(BC)$ is the tangent to Γ parallel to BC with Γ on the side of A. This tangent touches Γ at the midpoint W of the arc BC on the side opposite to A, so $h(D) = W$ and N, D, W are collinear. Arcs WB and WC are equal, so $\angle WBD = \angle WBC = \angle WNB$, and triangles WBD and WNB are similar. Hence $WB^2 = WD \cdot WN$, which is the power of W with respect to ω.

Step 2. Let V be the midpoint of DI_a, let X be the contact point of the A-excircle with BC, and let M be the midpoint of BC. By Lemma 5(2), M is the midpoint of DX. So $\overrightarrow{MV} = \frac{1}{2}\overrightarrow{XI_a}$, and V lies on the perpendicular bisector of BC, on the side opposite to A, with $MV = \frac{1}{2}r_a$. Put $\delta = MD$. By Pythagoras,

$$VB^2 = \tfrac{1}{4}a^2 + \tfrac{1}{4}r_a^2, \qquad VI^2 - r^2 = \delta^2 + \left(r + \tfrac{1}{2}r_a\right)^2 - r^2 = \delta^2 + rr_a + \tfrac{1}{4}r_a^2.$$

Now $\frac{1}{4}a^2 - \delta^2 = \left(\frac{1}{2}a - \delta\right)\left(\frac{1}{2}a + \delta\right) = BD \cdot DC$. Also $BI \perp BI_a$ gives $\angle IBD = 90^\circ - \angle XBI_a = \angle BI_aX$, so right triangles IDB and BXI_a are similar, and $rr_a = BD \cdot BX = BD \cdot DC$. Hence $VB^2 = VI^2 - r^2$, the power of V.

Step 3. For a point P, the quantity $PB^2 - (PI^2 - r^2)$ is a linear function of P (expand with vectors), so the points where the quantity vanishes form a line perpendicular to BI. This line is not parallel to the perpendicular bisector of BC, so exactly one point of the perpendicular bisector has PB^2 equal to its power. Both W and V are such points, so $W = V$. Thus N lies on line DI_a.

Step 4. Since $BD = BF$ and $DF \perp BI$, the external bisector BI_a is parallel to DF. So $I_a \notin DF$, and $N \neq F$ as N is on line DI_a. With directed angles, the tangent-chord angle at D and the isosceles triangle BDF give

$$\measuredangle(NF, NI_a) = \measuredangle(NF, ND) = \measuredangle(DF, DB) = \measuredangle(FB, FD) = \measuredangle(BF, BI_a),$$

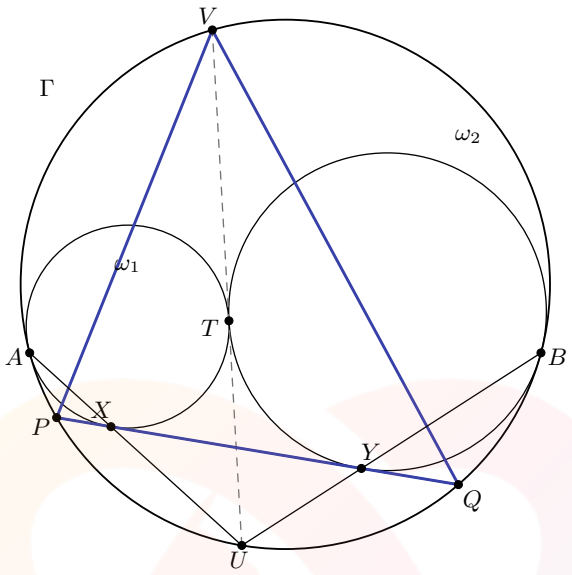
so B, F, N, I_a are concyclic. In the same way C, E, N, I_a are concyclic. ■

Remark. The guess $W = V$ comes from an accurate figure. Step 3 is the standard way to confirm a guessed point, as two conditions on W already pin W down.

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Problem 6

Circles ω_1 and ω_2 touch each other externally at T, and touch a circle Γ internally at A and B. A common external tangent of ω_1 and ω_2 meets Γ at P and Q. The common tangent of ω_1 and ω_2 at T meets Γ at U and V, where V is on the same side of line PQ as T. Prove that T is the incentre of triangle PQV.



Let line PQ touch ω_1 at X and ω_2 at Y. As PQ is an external tangent, ω_1 and ω_2 lie on the same side of PQ. Let U' be the midpoint of the arc PQ of Γ on the other side.

Step 1. The homothety at A taking ω_1 to Γ has ratio greater than 1, and sends line PQ to the tangent to Γ parallel to PQ with Γ on the side of ω_1. That tangent touches Γ at U'. So A, X, U' are collinear, with X between A and U', and X is inside Γ, so on chord PQ. In the same way B, Y, U' are collinear.

Step 2. Arcs $U'P$ and $U'Q$ are equal, so $\angle U'PX = \angle U'PQ = \angle U'AP$, and triangles $U'PX$ and $U'AP$ are similar. Hence $U'P^2 = U'X \cdot U'A$, the power of U' with respect to ω_1. Likewise $U'Q^2$ is the power of U' with respect to ω_2. As $U'P = U'Q$, the point U' lies on the radical axis of ω_1 and ω_2, which is their common tangent at T. So U' is U or V, and $U' \neq V$ because V is on the same side of PQ as T. Thus $U' = U$.

Step 3. Line UT touches ω_1 at T, so UT^2 is also the power of U with respect to ω_1. Hence $UT = UP = UQ$.

Step 4. In triangle PQV, the point U is the midpoint of arc PQ not containing V. By the incentre-excentre lemma, the incentre lies on segment VU at distance UP from U. The point T is inside Γ, so T lies on chord UV, and $UT = UP$. Hence T is the incentre of triangle PQV. ■

Remark. Step 2 shows that AX, BY and the common tangent at T are concurrent. The equality $UT = UP = UQ$ is what turns that concurrency into an incentre.

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